

Question 1

Question: How many applicant entries are for Fall 2026?

Final result: 33,207 applicants

SQL query:

```
SELECT COUNT(*)  
FROM applicants  
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

Explanation: This query determines the total number of records in the database associated with the Fall 2026 application term. The **WHERE** clause restricts the dataset before the count is performed, so only records whose **term** corresponds to Fall 2026 contribute to the result. **TRIM()** removes any accidental leading or trailing whitespace, while **LOWER()** standardizes capitalization, allowing values such as **Fall 2026** and **FALL 2026** to be treated consistently. After filtering, **COUNT(*)** counts every remaining applicant record. Therefore, the resulting **33,207** represents the number of Fall 2026 entries contained in this GradCafe dataset

Question 2

Question: Among applicants with a usable nationality classification, what percentage are international applicants?

Final result: 47.89%

SQL query:

```
SELECT
  100.0 *
  COUNT(*) FILTER (
    WHERE LOWER(TRIM(us_or_international))
      NOT IN ('american', 'other')
  )
  /
  NULLIF(
    COUNT(*) FILTER (
      WHERE us_or_international IS NOT NULL
        AND TRIM(us_or_international) <> ''
    ),
    0
  )
FROM applicants;
```

Explanation: This query calculates the percentage of applicants classified as international among records containing a usable nationality classification. The denominator counts records where `us_or_international` is neither NULL nor blank, preventing applicants with missing nationality information from affecting the percentage. For the numerator, `American` and `Other` are specifically excluded from the international category, consistent with the definition used for this analysis. The number of international records is divided by the total number of records with usable nationality information and multiplied by 100. `NULLIF(..., 0)` also protects the query from a division-by-zero error if no usable classifications existed. The resulting **47.89%** therefore describes the proportion of classified records in this dataset that meet the international definition used in the analysis.

Question 3

Question: What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores?

Final result:

Average GPA: 3.76

Average GRE Quantitative: 165.74

Average GRE Verbal: 160.37

Average GRE Analytical Writing: 4.33

SQL query:

```
SELECT
    AVG(gpa),
    AVG(gre) FILTER (
        WHERE gre BETWEEN 130 AND 170
    ),
    AVG(gre_v) FILTER (
        WHERE gre_v BETWEEN 130 AND 170
    ),
    AVG(gre_aw) FILTER (
        WHERE gre_aw BETWEEN 0.0 AND 6.0
    )
FROM applicants;
```

Explanation:

This query calculates four averages independently: GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing. This distinction is important because applicants frequently leave one or more test-score fields blank. PostgreSQL's **AVG()** function ignores NULL values, meaning an applicant who reported a GPA but no GRE score can still contribute to the GPA calculation without affecting any GRE calculation. For the GRE fields, **FILTER** clauses restrict Quantitative and Verbal scores to 130–170 and Analytical Writing scores to 0–6. During inspection of the database, some self-reported records contained clearly invalid values such as GRE scores of 999 and Analytical Writing values of 99.99; allowing those values into the calculation would substantially distort the averages. After excluding values outside the ranges used for these GRE fields, the resulting averages are **3.76 GPA, 165.74 GRE Quantitative, 160.37 GRE Verbal, and 4.33 GRE Analytical Writing.**

Question 4

Question: What is the average GPA of American applicants for Fall 2026?

Final result: 3.79

SQL query:

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND LOWER(TRIM(us_or_international)) = 'american'
      AND gpa IS NOT NULL;
```

Explanation:

This query examines a more specific subset of the applicant population by applying three conditions before calculating the average. First, the application term must be Fall 2026. Second, the applicant must be classified as **American** in the **us_or_international** field. Third, the applicant must have a non-NULL GPA. The **LOWER()** and **TRIM()** functions standardize the text fields so minor differences in capitalization or whitespace do not prevent otherwise matching records from being included. **AVG(gpa)** is then applied only to the records satisfying these conditions. The resulting **3.79** is therefore the mean reported GPA among American Fall 2026 applicants represented in the database who provided GPA information.

Question 5

Question: What percentage of Fall 2025 applicant entries are acceptances?

Final result: 40.74%

SQL query:

```
SELECT
  100.0 *
  COUNT(*) FILTER (
    WHERE LOWER(TRIM(status)) LIKE 'accept%'
  )
  /
  NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

Explanation:

This query measures what proportion of the Fall 2025 records in the database have an acceptance status. The outer **WHERE** clause first restricts the population to Fall 2025, which establishes the denominator for the calculation. Within that population, **COUNT(*) FILTER (...)** counts records whose normalized status begins with **accept**. Using **LIKE 'accept%'** allows the query to recognize acceptance-related status text without requiring one exact capitalization or string representation. The number of accepted entries is divided by the total number of Fall 2025 entries and multiplied by 100. **NULLIF(COUNT(*), 0)** prevents division by zero if no Fall 2025 records existed. The resulting **40.74%** is therefore the acceptance percentage among the self-reported Fall 2025 entries contained in this database, rather than an institutional or national graduate-school acceptance rate.

Question 6

Question: What is the average GPA of accepted Fall 2026 applicants?

Final result: 3.76

SQL query:

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND LOWER(TRIM(status)) LIKE 'accept%'
      AND gpa IS NOT NULL;
```

Explanation:

This query combines term, admission status, and GPA availability to examine the academic profile of accepted Fall 2026 applicants. The first condition restricts the records to Fall 2026, while the second keeps only records whose status begins with **accept**. The final condition requires a reported GPA so that missing values do not contribute to the calculation. **AVG(gpa)** then computes the arithmetic mean across the qualifying GPA values. The resulting **3.76** represents the average reported GPA of accepted Fall 2026 applicants in this GradCafe sample who supplied GPA information. Because the database consists of self-reported entries, this value describes the observed sample and should not be interpreted as the average GPA of all graduate students admitted nationally in Fall 2026.

Question 7

Question: How many Johns Hopkins University master's-level Computer Science applicant entries are in the database?

Final result: 18 applicants

SQL query:

```
SELECT COUNT(*)
FROM applicants
WHERE (
    LOWER(program) LIKE '%johns hopkins%'
    OR LOWER(program) LIKE '%jhu%'
)
AND LOWER(program) LIKE '%computer science%'
AND (
    LOWER(TRIM(degree)) IN (
        'ms',
        'm.s.',
        'msc',
        'm.sc.',
        'master',
        'masters',
        'master"s'
    )
    OR LOWER(degree) LIKE '%master%'
);
```

Explanation:

This query identifies records that simultaneously satisfy university, program, and degree-level requirements. Because the database's **program** field contains the original university and program information together, the first condition searches that field for either **Johns Hopkins** or the abbreviation **JHU**. A second condition requires the same field to contain **Computer Science**, preventing records from other Johns Hopkins programs from being counted. The degree condition accounts for several possible ways applicants may have entered a master's degree, including **MS**, **M.S.**, **MSc**, **Master**, and related variations. The broader **LIKE '%master%'** condition captures additional master's labels that may not exactly match one of the listed strings. Only records satisfying all of these conditions are counted, resulting in **18 Johns Hopkins master's-level Computer Science entries** in the database.

Question 8

Question: Using the original downloaded fields, how many accepted Fall 2026 Computer Science PhD applicant entries are from Georgetown University, MIT, Stanford University, or Carnegie Mellon University?

Final result: 30 applicants

SQL query:

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accept%'
AND (
    LOWER(TRIM(degree)) IN (
        'phd',
        'ph.d.',
        'ph.d'
    )
    OR LOWER(degree) LIKE '%doctor%'
)
AND LOWER(program) LIKE '%computer science%'
AND (
    LOWER(program) LIKE '%georgetown%'
    OR LOWER(program) LIKE
        '%massachusetts institute of technology%'
    OR LOWER(program) LIKE '%mit%'
    OR LOWER(program) LIKE '%stanford%'
    OR LOWER(program) LIKE '%carnegie mellon%'
);
```

Explanation:

This query uses the original downloaded program information to identify a narrowly defined group: accepted Fall 2026 Computer Science PhD applicants at Georgetown, MIT, Stanford, or Carnegie Mellon. The first condition restricts the term to Fall 2026, and the second requires an acceptance status. The degree conditions accommodate multiple representations of a doctorate, such as **PhD**, **Ph.D.**, and text containing **doctor**. The query then requires the original combined **program** field to contain **computer science**. Finally, that same original field is searched for the names or recognized forms of the four specified universities. Since all of these conditions are

connected with **AND**, a record must satisfy the term, acceptance, degree, program, and university requirements simultaneously. The resulting original-field count is **30 records**.

Question 9

Question: When the university and program are identified using the LLM-generated fields instead of the original downloaded fields, how many accepted Fall 2026 Computer Science PhD entries are found at Georgetown, MIT, Stanford, and Carnegie Mellon, and how does this compare with Question 8?

Final result:

Original-field count: 30

LLM-field count: 30

Difference: +0

SQL query:

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accept%'
AND (
    LOWER(TRIM(degree)) IN (
        'phd',
        'ph.d.',
        'ph.d'
    )
    OR LOWER(degree) LIKE '%doctor%'
)
AND LOWER(llm_generated_program)
LIKE '%computer science%'
AND (
    LOWER(llm_generated_university)
    LIKE '%georgetown%'
    OR LOWER(llm_generated_university)
    LIKE '%massachusetts institute of technology%'
    OR LOWER(llm_generated_university) = 'mit'
    OR LOWER(llm_generated_university)
    LIKE '%stanford%'
    OR LOWER(llm_generated_university)
    LIKE '%carnegie mellon%'
);
```

Explanation:

This query repeats the analysis performed in Question 8 but changes how the university and academic program are identified. Instead of searching the original combined **program** field for both pieces of information, it searches **llm_generated_program** for Computer Science and **llm_generated_university** for Georgetown, MIT, Stanford, or Carnegie Mellon. The original **term**, **status**, and **degree** fields are retained so that the main difference between Questions 8 and 9 is the source used to identify the program and university. This provides a direct comparison between the original downloaded text and the standardized fields generated during the LLM data-processing step. The original-field query returned **30 records**, while the LLM-field query also returned **30 records**, producing a difference of **0**. Therefore, for this particular subset of the data, using the LLM-generated university and program fields did not change the final number of qualifying records. This does not necessarily mean the two approaches produce identical classifications throughout the entire dataset; it only shows that they produced the same aggregate count for this specific analysis.

Question 10 (Original Questions)

Question: Among Princeton University applicant entries for Fall 2026, what percentage are acceptances?

Final result: 20.77%

SQL query:

```
SELECT
  100.0 *
  COUNT(*) FILTER (
    WHERE LOWER(TRIM(status)) LIKE 'accept%'
  )
  /
  NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(program) LIKE '%princeton%';
```

Explanation:

This query investigates Princeton University specifically by calculating the percentage of its Fall 2026 records that have an acceptance status. The **WHERE** clause first restricts the dataset to Fall 2026 and then searches the original combined program information for **princeton**. These qualifying records form the denominator. Within this Princeton subset, **COUNT(*) FILTER (...)** counts records whose status begins with **accept**, creating the numerator. Dividing the accepted-record count by the total Princeton Fall 2026 record count and multiplying by 100 produces **20.77%**. This result must be interpreted specifically as the acceptance percentage among Princeton Fall 2026 entries contained in the scraped, self-reported GradCafe dataset. It is **not Princeton University's official graduate acceptance rate**, because people voluntarily submitting results to GradCafe are not necessarily representative of the entire Princeton applicant population.

Question 11 (Original Questions)

Question: What is the average GPA of accepted Princeton University applicants for Fall 2026 who reported a GPA?

Final result: 3.87

SQL query:

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
  AND LOWER(program) LIKE '%princeton%'
  AND LOWER(TRIM(status)) LIKE 'accept%'
  AND gpa IS NOT NULL;
```

Explanation:

This query further analyzes the Princeton Fall 2026 subset by examining the reported GPAs of accepted applicants. It first requires the term to be Fall 2026 and searches the original combined program field for **princeton**. It then restricts the records to statuses beginning with **accept** and requires **gpa IS NOT NULL**, ensuring that only applicants who actually supplied GPA information contribute to the calculation. **AVG(gpa)** computes the mean of those qualifying GPA values, producing an average of **3.87**. The result therefore represents the average reported GPA among accepted Princeton Fall 2026 applicants found in this GradCafe sample who provided a GPA. It should not be interpreted as Princeton's official average admitted-student GPA because the underlying dataset is anonymous, voluntarily submitted, and potentially affected by missing data and selection bias.